

Introduction to CCR and xVA

Advanced Financial Modeling

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Suggested References

Everything will be covered during the course and provided to you as slides, notes or Python notebooks.

If you want more, check the following:

- D. Brigo, F. Mercurio (2006) Interest Rate Models - Theory and Practice (2nd edition).
- J. Gregory (2020) The xVA Challenge.

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2 Exposure Generation

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Intro

Consider a trade between two counterparties A and B.

Counterparty A enters a payer interest rate swap, receiving the floating rate:

$$\text{IRS}(t) = \sum_{k=1}^N \tau_k P(t, T_k) F(t, T_{k-1}, T_k) - K \sum_{k=1}^N \tau_k P(t, T_k)$$

Assume that right before one of the payments there is a shock leading to a monetary tightening.

Intro

In this new market scenario, the increase in interest rates makes the position more valuable for counterparty A.

- Counterparty A has a positive exposure to B
- Counterparty B may default and fail to meet its obligations
- Counterparty A may incur losses

This risk is known as Counterparty Credit Risk (CCR).

Counterparty Credit Risk

- CCR is the risk that a counterparty defaults before contract maturity
- The future exposure is uncertain and depends on market conditions
- A contract with default risk is less valuable than a risk-free one

$$V^{\text{risky}}(t) = V^{\text{risk-free}}(t) - CVA(t)$$

Credit Adjustments: CVA and DVA

Credit Value Adjustment (CVA) represents the expected loss due to counterparty default.

- It reduces the value of the contract
- It depends on exposure, default probability and recovery

A simple approximation is:

$$\mathbf{uCVA} \approx LGD_{\text{ctp}} \times PD_{\text{ctp}} \times EPE$$

Debt Value Adjustment (DVA) reflects the impact of the institution's own default risk:

$$\mathbf{uDVA} \approx LGD_{\text{self}} \times PD_{\text{self}} \times ENE$$

CVA: Continuous-Time Representation

$$\text{uCVA}(t) = \mathbb{E}^Q \left[1_{t < \tau \leq T} \cdot LGD(\tau) \cdot D(t, \tau) \cdot PV^+(\tau) \mid \mathcal{F}_t \right]$$

- $PV^+(\tau)$: positive exposure at default
- $D(t, \tau)$: discount factor
- LGD : loss given default
- τ : default time

CVA under Intensity-Based Models

Assuming independence between exposure and default:

$$\text{uCVA}(t) = \int_t^T \mathbb{E}^Q [LGD(u) D(t, u) PV^+(u)] d\mathbb{Q}(\tau \leq u)$$

Using the default intensity $\lambda(u)$:

$$\text{uCVA}(t) = \int_t^T \mathbb{E}^Q [LGD(u) D(t, u) PV^+(u)] \lambda(u) e^{-\int_t^u \lambda(s) ds} du$$

Discrete Approximation

In practice, CVA is computed on a time grid:

$$\text{uCVA}(t) \approx \sum_{i=1}^n \mathbb{E}^Q [LGD(t_i) D(t, t_i) PV^+(t_i)] \cdot \Delta PD(t_i)$$

- $\Delta PD(t_i) = \mathbb{Q}(t_{i-1} < \tau \leq t_i)$
- Under intensity models:

$$\Delta PD(t_i) \approx \lambda(t_i) e^{-\int_t^{t_i} \lambda(s) ds} \Delta t$$

- It represents the default probability over each time interval

Funding Value Adjustment (FVA)

Even in absence of default, exposures need to be funded.

Positive Exposure ($PV^+ > 0$):

- The contract has positive value but no immediate cash inflow
- The bank must borrow funds to finance this position
- $\Rightarrow$ Funding Cost Adjustment (FCA)

Negative Exposure ($PV^- < 0$):

- The bank is effectively holding cash until future payments
- This surplus can be invested
- $\Rightarrow$ Funding Benefit Adjustment (FBA)

$$\mathbf{FVA = FBA - FCA}$$

Exposure, Collateral and Funding: Key Mechanism

Key insight: A positive mark-to-market is not the same as holding cash.

Case 1: Perfect collateralization

- If $PV(t) > 0$, the counterparty posts collateral
- The bank receives cash and can reuse it
- $\Rightarrow$ No funding need $\Rightarrow FVA \approx 0$

Case 2: Imperfect or no collateral

- Exposure is only partially covered (thresholds, delays, MTA)
- The bank holds a positive asset without immediate cash
- $\Rightarrow$ External funding is required $\Rightarrow FCA$ arises

Conclusion:

- Funding depends on *cash flows*.
- Collateralization is the main driver of funding needs
- FVA is model and institution-dependent

FVA: Valuation

$$\begin{aligned} \text{FVA}(t) = & \sum_{i=1}^n f_b(t_i) \tau_i \mathbb{E}^Q [S_C(t, t_{i-1}) S_B(t, t_{i-1}) D(t, t_i) \text{PV}(t_i)^+] \\ & - \sum_{i=1}^n f_l(t_i) \tau_i \mathbb{E}^Q [S_C(t, t_{i-1}) S_B(t, t_{i-1}) D(t, t_i) \text{PV}(t_i)^-] \end{aligned}$$

- S_B, S_C : survival probabilities of bank and counterparty
- Funding costs arise only while both parties are alive
- If either defaults, the contract terminates and funding stops

Credit and Funding Analogy

Positive Exposure:

- Generates counterparty credit risk (CVA)
- Is also the amount that has to be funded when the counterparty does not default (FCA)

Negative Exposure:

- Reflects own default risk (DVA)
- Generates funding benefit (FBA)

Two common framework to avoid a double-counting of funding benefits:

$$V^D(t) \approx V(t) - [CVA(t) - DVA(t)] - FCA(t)$$

$$V^D(t) \approx V(t) - CVA(t) + [FBA(t) - FCA(t)]$$

Remarks

- Pricing under counterparty credit risk requires modelling the distribution of future exposures, captured through XVA adjustments
- The most computationally demanding component is the estimation of Expected Exposure and its sensitivities
- Exposure modelling is complex due to:
 - long time horizons
 - multiple risk factors
 - risk mitigants such as netting and collateral
- Cross-asset modelling is required, often relying on simplifying assumptions for default probabilities and recovery rates
- **Key challenge:** balancing model realism and computational tractability

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Pricing Exposures

We define the exposure of a derivative $E(t)$, the positive exposure $PE(t)$ and the negative exposure $NE(t)$ at some instant t as

$$E(t) = V(t)$$

$$PE(t) = \max(V(t), 0)$$

$$NE(t) = \min(V(t), 0)$$

the expectations in t_0 , under $\mathbb{Q}$ -measure, are given by

$$EE(t_0, t) = \mathbb{E}^{\mathbb{Q}} \left[\frac{N(t_0)}{N(t)} E(t) | \mathcal{F}(t_0) \right],$$

$$EPE(t_0, t) = \mathbb{E}^{\mathbb{Q}} \left[\frac{N(t_0)}{N(t)} PE(t) | \mathcal{F}(t_0) \right],$$

$$ENE(t_0, t) = \mathbb{E}^{\mathbb{Q}} \left[\frac{N(t_0)}{N(t)} NE(t) | \mathcal{F}(t_0) \right],$$

Pricing Exposures

xVA as Exotic Options

When default risk is included in the risk-neutral valuation, the payoff depends both on the underlying of the derivative and on the default event.

This transforms quantities such as $PE(t)$ and $NE(t)$ into *credit-hybrid, path-dependent payoffs*, similar to exotic options.

- A key driver is the volatility of the underlying risk factors, which directly affects future exposures
- Margining and collateral introduce additional path-dependence
- Risk-neutral valuation and sensitivity computation are therefore computationally demanding
- Default introduces a stopping time, making the valuation similar to barrier or American-style features

Simulation of Exposure

Exposure modelling is essentially an option-pricing problem.

The most common and flexible approach is Monte Carlo simulation.

Main challenges:

- **Model choice:** selecting appropriate dynamics for multiple risk factors, often across different asset classes
- **Calibration:** fitting models to market and historical data is challenging due to the high dimensionality
- **Numerical implementation:** discretization schemes and simulation techniques affect both accuracy and efficiency
- **Computational resources:** large-scale simulations and sensitivities require significant computational power

Key insight: XVA is a high-dimensional, simulation-based problem



IRS Exposure

Consider a vanilla interest rate swap, where two counterparties exchange fixed and floating cash flows.

For a payer swap (pay fixed, receive floating), under a single-curve framework and assuming common payment dates, the value can be written as:

$$V(t) = N \sum_{k=1}^m \tau_k P(t, T_k) (L(t; T_{k-1}, T_k) - K) \quad (1)$$

where:

- N is the notional
- $T_1, \dots, T_m$ are the payment dates
- $\tau_k = T_k - T_{k-1}$
- $L(t; T_{k-1}, T_k)$ is the forward rate

IRS Exposure

We model the short rate using the one-factor Hull–White model:

$$dr(t) = (\theta(t) - \kappa r(t))dt + \sigma dW(t)$$

The conditional distribution of $r(t)$ is Gaussian with:

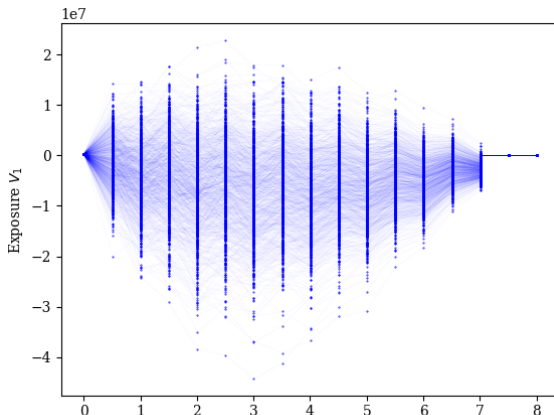
$$\begin{aligned}\mathbb{E}[r(t) \mid r(s)] &= r(s)e^{-\kappa(t-s)} + \int_s^t e^{-\kappa(t-u)}\theta(u) du, \\ \text{Var}[r(t) \mid r(s)] &= \frac{\sigma^2}{2\kappa} \left(1 - e^{-2\kappa(t-s)}\right).\end{aligned}$$

Zero-coupon bond prices are given by:

$$P(t, T) = \mathbb{E}^Q \left[e^{-\int_t^T r(u) du} \mid \mathcal{F}_t \right]$$

IRS Exposure

We are interested in $\sum_{i=1}^m ENE(t, t_i)$ and $\sum_{i=1}^m EPE(t, t_i)$ along the time grid discretization m (V_1 receiver swap with 7y maturity and fair strike)



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Netting

Netting is a common mechanism used to reduce counterparty risk when multiple transactions generate both positive and negative exposures.

- **Single counterparty:** *bilateral netting* offsets positive and negative exposures across different trades within the same netting set for one counterparty.
- **Multiple counterparties:** *multilateral netting* offsets exposures across several counterparties within a centralized arrangement, such as a clearing house.

Netting

A portfolio may contain many transactions whose exposures partially offset each other.

In general, netting is a mechanism for aggregating obligations that preserves the underlying market risk profile while reducing:

- settlement risk,
- counterparty risk,
- operational risk,
- liquidity risk,
- systemic risk.

Netting

For a netting set containing n trades, the exposure, positive exposure, and negative exposure at time t are defined as

$$E_{ns}(t) = \sum_{i=1}^n V_i(t),$$

$$PE_{ns}(t) = \max\{E_{ns}(t), 0\},$$

$$NE_{ns}(t) = \min\{E_{ns}(t), 0\}.$$

Netting

Jensen Inequality

$\max(0, x)$ is a convex function, hence for a sample of T elements

$$\max\left(0, \frac{1}{T} \sum_{i=1}^T x_i\right) \leq \frac{1}{T} \sum_{i=1}^T \max(0, x_i)$$

which immediately leads, for a specific netting set, container of n trades, to

$$|CVA_{ns}(t)| \leq \left| \sum_{i=1}^n CVA_i \right|$$

Collateral

- Future exposure can grow to a large and potentially unmanageable level.
- A common way to mitigate this risk is to embed contractual provisions that trigger actions aimed at reducing exposure.
- In exchange-traded markets, margin posting works essentially as a daily settlement mechanism.
- In over-the-counter (OTC) derivatives, by contrast, it operates through a collateralisation process.

Collateral

Two main types of margin can be distinguished:

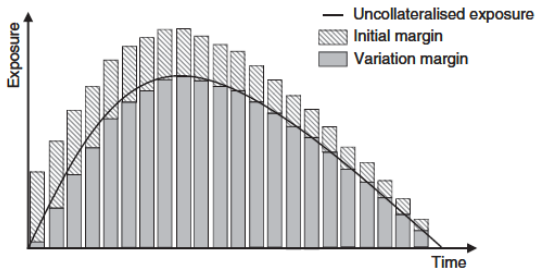
- **Initial Margin:** also referred to as the *independent amount*. It is not netted against variation margin and may be segregated.
- **Variation Margin:** reflects current exposure and is defined in the CSA through the credit support amount.

If C denotes the collateral amount, then the exposure at time t is

$$\begin{aligned}E(t) &= V(t) - C, \\PE(t) &= \max(V(t) - C, 0), \\NE(t) &= \min(V(t) - C, 0).\end{aligned}$$

Collateral

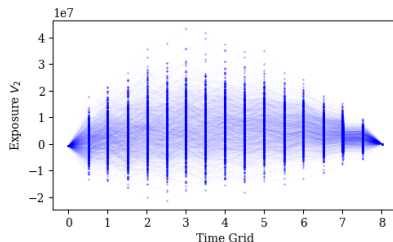
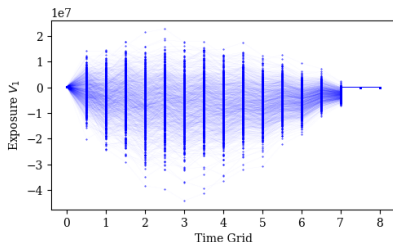
Impact of margin on a typical exposure profile



source: J. Gregory (2020) *The xVA Challenge*.

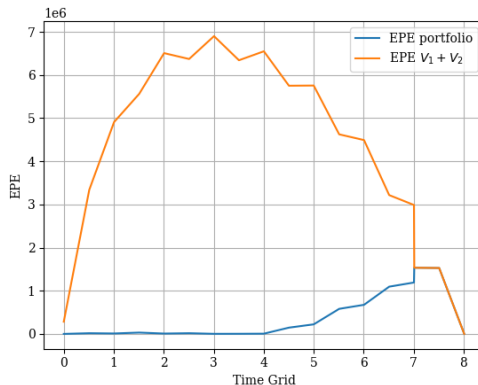
IRS Netting-Portfolio

Let V_1 be the previously defined receiver swap, and let V_2 be a payer swap, with 80M notional and 8y maturity:



IRS Netting-Portfolio

The two contracts offset (hedge) each other exposure as they are written on the same underlying, they have similar size, strike and maturity



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Poisson Process

Time-Homogeneous Poisson Process

A time-homogeneous Poisson process $\{M_t, t \geq 0\}$ is a right-continuous counting process with unit jumps, independent and stationary increments, and initial condition $M_0 = 0$.

These assumptions imply that the first jump time τ^1 is exponentially distributed. In particular, there exists a constant intensity $\lambda > 0$ such that

$$\mathbb{Q}(M_t = 0) = \mathbb{Q}(\tau^1 > t) = e^{-\lambda t}.$$

Poisson Process

Time-Inhomogeneous Poisson Process

More generally, the jump intensity may vary over time. Conditional on no jump before t , the risk-neutral probability of a jump in the next interval $[t, t + dt]$ is

$$\mathbb{Q}(\tau \in [t, t + dt] \mid \tau > t) = \lambda(t) dt.$$

Here $\lambda(t) > 0$ is called the **intensity** or **hazard rate**.

We also define the cumulative intensity

$$\Lambda(t) = \int_0^t \lambda(u) du,$$

also known as the **hazard function**.

Survival Probability

From Intensity to Survival Probability

For a deterministic intensity, the survival probability is

$$\mathbb{Q}(\tau > t) = e^{-\int_0^t \lambda(u) du}.$$

Hence, the default probability over $[0, t]$ is

$$\mathbb{Q}(\tau \leq t) = 1 - e^{-\int_0^t \lambda(u) du}.$$

Intensity and Discount Factors

Analogy with Interest Rates

The survival probability can be written as

$$\mathbb{Q}(\tau > t) = e^{-\int_0^t \lambda(u) du},$$

which has exactly the same form as the price of a zero-coupon bond in an interest rate model:

$$P(0, t) = e^{-\int_0^t r(u) du}.$$

Therefore:

- interest rates generate **discount factors**,
- default intensities generate **survival probabilities**.

Stochastic Intensity

Stochastic Hazard Rate

If the intensity is stochastic, the survival probability becomes

$$\mathbb{Q}(\tau > t) = \mathbb{E} \left[e^{-\int_0^t \lambda(u) du} \right].$$

This is again analogous to the price of a zero-coupon bond in a short-rate model, with the short rate r replaced by the default intensity λ .

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Managing xVA

- An xVA portfolio inherits sensitivities to all market factors driving the underlying transactions, across currencies, asset classes, and product types.
- In addition, xVA generates specific sensitivities of its own; for example, volatility risk may arise even for products that are not explicitly volatility-sensitive.
- From a risk-management perspective, xVA can be viewed much like an options portfolio. For example, the exposure profile of a swap may be interpreted as a strip of European swaptions.

Managing xVA

Where is the risk?

$$dxVA \approx \frac{\partial xVA}{\partial S} dS + \frac{\partial xVA}{\partial E} dE + \frac{\partial xVA}{\partial t} dt + \frac{\partial^2 xVA}{\partial S \partial E} dS dE$$

Here, dE represents movements in exposure drivers, including:

- interest rate risk, including multi-curve and basis effects;
- interest rate volatility risk;
- FX spot risk, cross-currency basis risk, and FX volatility risk.

By contrast, dS represents credit spread movements, and the interaction term captures cross-gamma effects between market and credit factors.

Cross-Gamma and Wrong-Way Risk

- **Wrong-way risk** occurs when exposure is high exactly in states where the counterparty is more likely to default.
- **Cross-gamma** captures the effect of joint moves in two risk factors. Hedging each risk separately does not, in general, eliminate the impact of their interaction.
- These effects are often difficult to hedge and can generate substantial P&L volatility.
- For example, a fall in interest rates combined with a widening of credit spreads may significantly affect xVA.
- Modeling this dependence generally requires stochastic default intensities.

Monte Carlo Sensitivities

xVA is typically exposed to a large number of market risk factors.

- Estimating all xVA sensitivities by finite differences requires rerunning the Monte Carlo simulation many times, with each input bumped separately.
- Such a brute-force approach is usually impractical, due to its computational cost and the associated hardware, implementation, and maintenance burden.
- The alternative has often been to use simplified approximations, trading off accuracy for speed.
- A more efficient solution is provided by **Adjoint Algorithmic Differentiation (AAD)**, which allows many sensitivities to be computed simultaneously at a cost largely independent of the number of input risk factors.

xVA Wrap-Up

xVA adjustments arise because derivative valuation in practice must reflect counterparty risk, funding, collateral, capital, and margin costs.

As a result, pricing is no longer based only on risk-free discounted cash flows, but on the joint evolution of **market factors, exposure, credit quality, and balance-sheet constraints**.

The key practical insight is that xVA is inherently a **portfolio-level, dynamic, and simulation-based** problem.

Hence, modern xVA desks rely on netting, collateral management, Monte Carlo exposure models, and efficient sensitivity techniques such as **AAD**.